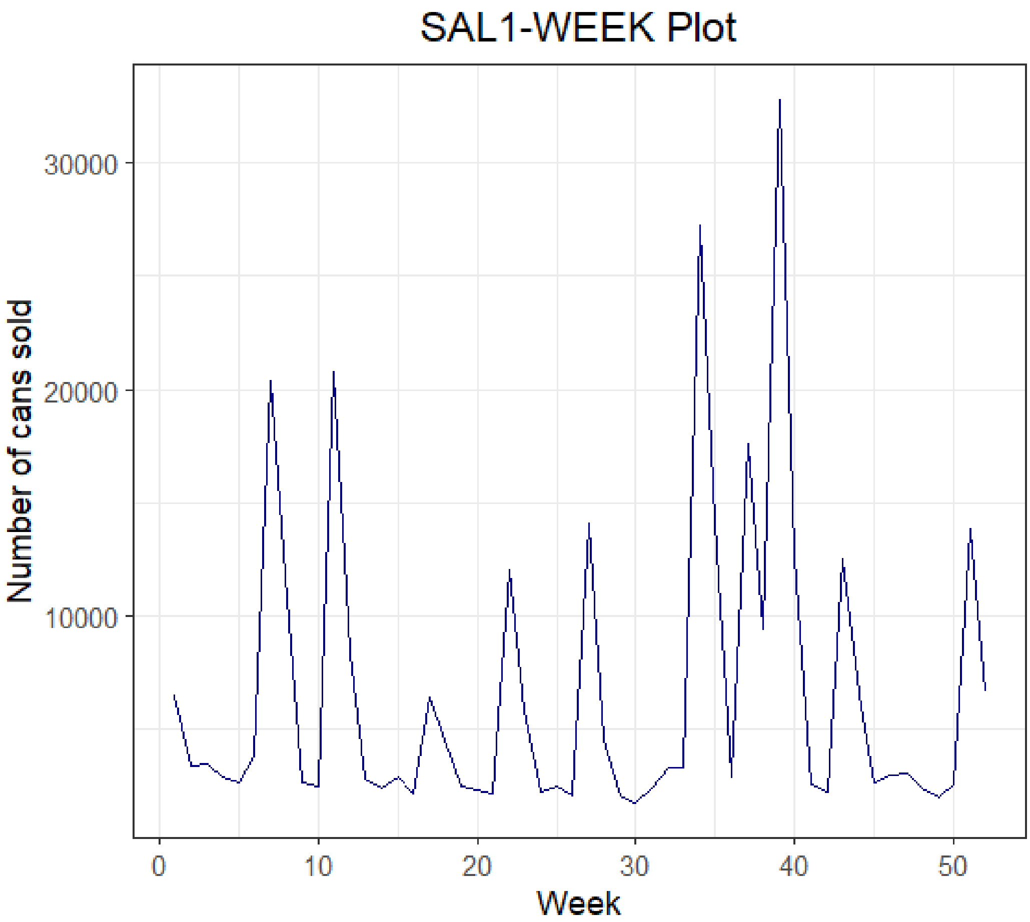
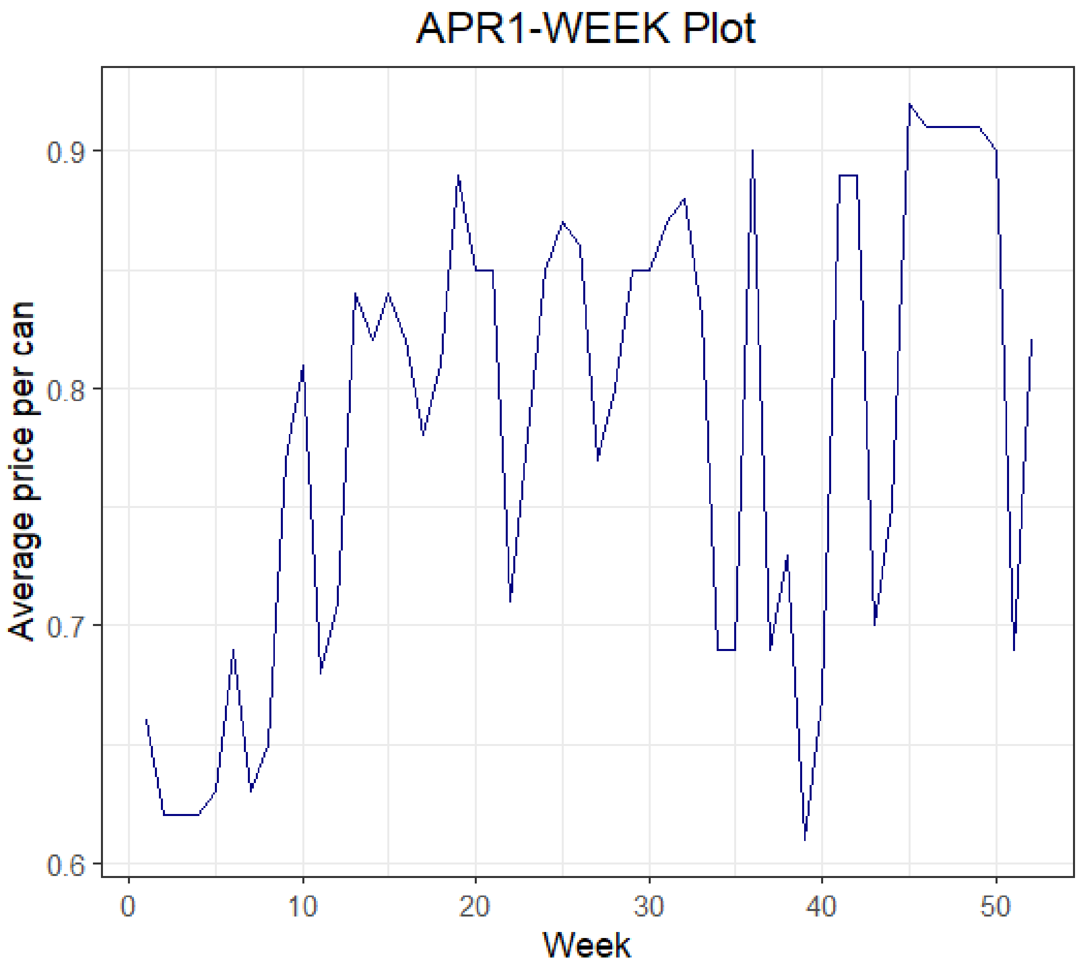
a. Calculate the summary statistics for *SAL*1 and *APR*1. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?

```
#(a.)
install.packages("ggplot2")
rm(list=ls())
library(ggplot2)
library(POE5Rdata)

data("tuna")

tuna <- cbind(week = 1:nrow(tuna), tuna)

summary_stats <- data.frame(
  Variable = c("SAL1", "APR1"),
  Mean = c(mean(tuna$sal1), mean(tuna$apr1)),
  Min = c(min(tuna$sal1), min(tuna$apr1)),
  Max = c(max(tuna$sal1), max(tuna$apr1)),
  SD = c(sd(tuna$sal1), sd(tuna$apr1))
)
print(summary_stats)


ggplot(tuna, aes(x = week, y = sal1)) +
  geom_line(color = "navy") +
  scale_y_continuous(breaks = seq(0, 40000, by = 10000)) +
  labs(title = "SAL1-WEEK Plot",
       x = "Week",
       y = "Number of cans sold") +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5))

ggplot(tuna, aes(x = week, y = apr1)) +
  geom_line(color = "navy") +
  scale_y_continuous(breaks = seq(0, 1, by = 0.1)) +
  labs(title = "APR1-WEEK Plot",
       x = "Week",
       y = "Average price per can") +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5))
```

The summary statistics for SAL1 and APR1 is below:

	Variable	Mean	Min	Max	SD
1	SAL1	6718.7115	1762.00	32820.00	6.915083e+03
2	APR1	0.7825	0.61	0.92	9.803711e-02

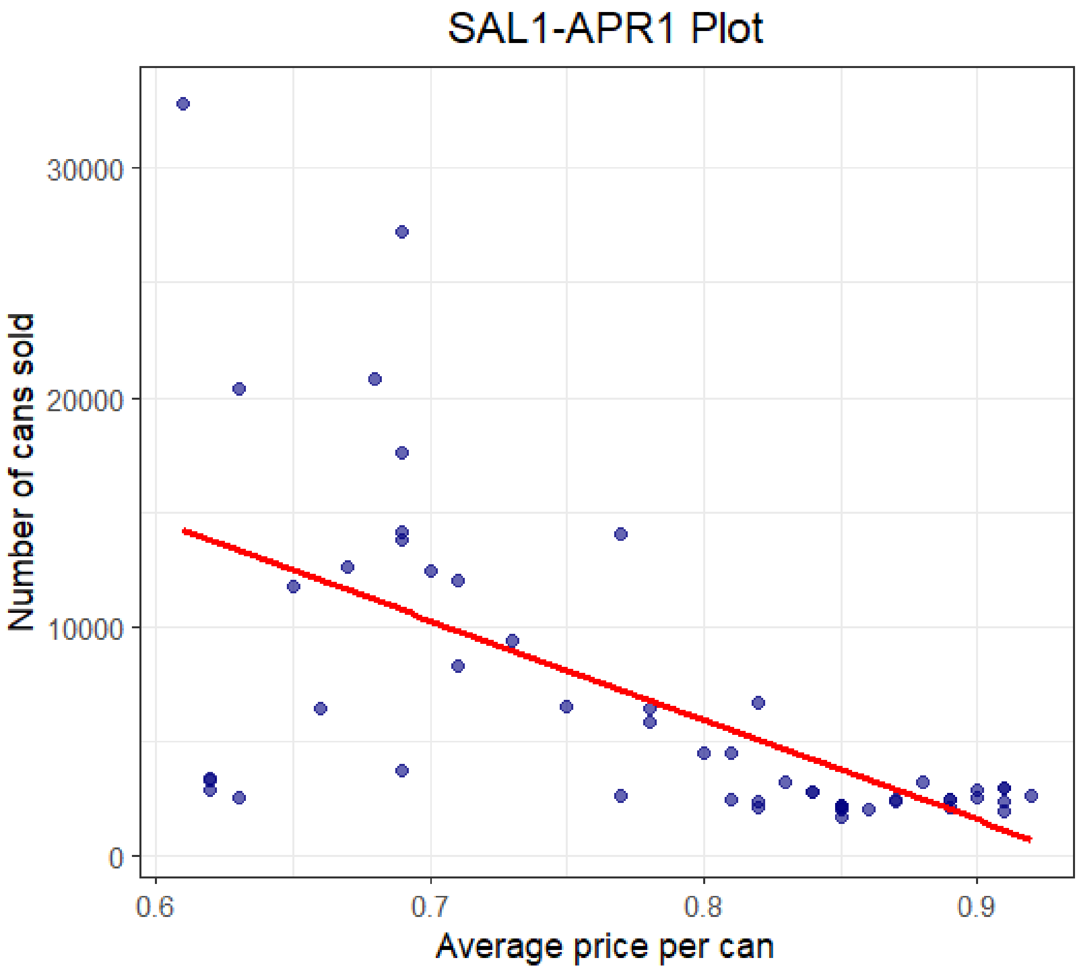




Conclusion: As you can see this two graphs, weekly sales exhibit much greater variation than weekly price.

b. Plot the variable *SAL*1 (*y*-axis) against *APR*1 (*x*-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

```
#(b.)
ggplot(tuna, aes(x = apr1, y = sal1)) +
  geom_point(color = "navy", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(title = "SAL1-APR1 Plot",
       x = "Average price per can",
       y = "Number of cans sold") +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5))
```



Conclusion: There is an inverse relationship between the weekly sales and weekly price, and it shows that the higher the price, the lower the sales volume, which is what we expected.

c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

```
#(c)
price1 = 100 * tuna$apr1
mod1 = lm(tuna$sal1 ~ price1)
b1 = coef(mod1)[1]
b2 = coef(mod1)[2]  #加1美分銷量減少b2罐

alpha = 0.05
df = df.residual(mod1)              #degree of freedom
tc = qt(1-alpha/2, df)                #critical points
smod1 = summary(mod1)
se_b2 = coef(smod1)[2,2]
lowb = b2 - tc*se_b2
upb  = b2 + tc*se_b2

cat(
  "3.31(c)", "\n",
  "a one-cent increase in the price of brand one will reduce the expected", "\n",
  "weekly sales of brand one tuna by ",-b2," cans", "\n",
  "and 95% CI estimate is [",lowb,",",upb,"]","\n",
  sep=""
```

Conclusion: A one-cent increase in the price of brand one will reduce the expected weekly sales of brand one tuna by 434.4473 cans, which is the point estimate. The 95% CI estimate is [-592.2897,-276.6049].

d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

```
#(d)
# price1 = 70
e_hat = b1 + b2 * 70

alpha_d = 0.1
tc_d = qt(1-alpha_d/2, df)                    #critical points
se_l = sqrt(vcov(mod1)[1,1] + 4900*vcov(mod1)[2, 2] + 140*vcov(mod1)[1,2])

lowb_d = e_hat - tc_d*se_l
upb_d  = e_hat + tc_d*se_l
```

Conclusion: The 90% CI estimate is [8624.934,11980.87].

e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

```r
#(e)
mod2 = lm(tuna$sal1 ~ tuna$apr1)
b2_e = coef(mod2)[2]
smod2 = summary(mod2)

alpha_e = 0.05
c = mean(tuna$apr1) / mean(tuna$sal1)
ela_hat = b2_e * c
se_ela = coef(smod2)[2,2] * abs(c)
tc_e = qt(1-alpha_e/2, df)          #自由度不變所以沒改

lowb_e = ela_hat - tc_e*se_ela
upb_e  = ela_hat + tc_e*se_ela
```

```r
cat(
  "3.31(e)", "\n",
  "estimate of the elasticity is ",ela_hat, "\n",
  "The 95% CI estimate is [",lowb_e,",",upb_e,"]","\n",
  "since absolute value of elasticity is ",abs(ela_hat), "\n",
  "this is elastic and hence the result satisfied Law of Demand and Supply", "\n",
  sep=""
)
```

Conclusion:

Estimate of the elasticity is -5.059825

The 95% CI estimate is [-6.898149,-3.221501]

Since absolute value of elasticity is 5.059825, this is elastic and hence the result satisfied Law of Demand and Supply.

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the *p*-value for the test. What is your conclusion?

```r
#(f)
# test statistic is (ela_hat + 3)/ (c *se(b2))
alpha_e = 0.1
t_test = (ela_hat + 3) / se_ela
tc_f = qt(1-alpha_e/2, df)

# state the null and alternative hypotheses in terms of the model parameters
# 這邊會另外寫
# h0 = 3,h1 != 3

cat(
  "3.31(f)", "\n",
  "the rejection region is tc > ",tc_f," or tc < ",-tc_f, "\n",
  "and test p-value is ",t_test, "\n",
  "hence We reject the null hypothesis", "\n",
  sep=""
)
```

Conclusion:

The rejection region is tc > 1.675905 or tc < -1.675905

The test t-value is -2.250572

The p-value is 0.02884204, which is smaller than 0.5 * alpha = 0.05

Consequently, we reject the null hypothesis.